Quantification: 701_z19c1-2
==========================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
----

measure	value
lacuna count	31
interior lacuna count	23
frame-edge lacuna count	8

Interior lacunae only: mean, median and sample SD (n = 23)
----

measure	kind	mean	median	sd	units
area_px2	lacuna shape	2804.39	1270	3446.05	px^2
major_axis_length_px	lacuna shape	80.3542	71.948	41.6335	px
minor_axis_length_px	lacuna shape	43.9972	33.0049	31.6942	px
aspect_ratio	lacuna shape	2.0963	2.089	0.7425	unitless
eccentricity	lacuna shape	0.8047	0.878	0.1673	unitless
solidity	lacuna shape	0.7461	0.7246	0.0922	unitless
roots_count	headline	10.087	8	6.2225	count
ring_length_r30_px	headline	368.652	322	200.205	px
ring_length_r60_px	reported	729.609	614	367.946	px
owned_length_px	ownership-dependent	802.996	705.784	592.588	px
edge_count	ownership-dependent	51.913	43	38.6463	count
mean_edge_length_px	ownership-dependent	15.8531	15.4393	3.5499	px
perimeter_px	normalised	358.137	255.806	303.693	px
ring_area_r30_px2	normalised	8259.7	6964	4578.36	px^2
ring_area_r60_px2	normalised	16213.2	16328	7818.81	px^2
in_frame_fraction_r30	normalised	0.9905	1	0.0453	unitless
in_frame_fraction_r60	normalised	0.9821	1	0.0687	unitless
ring_density_r30	normalised	0.044887	0.0423219	0.0133165	px^-1
ring_density_r60	normalised	0.0447087	0.0435629	0.0107539	px^-1
roots_per_100px_perimeter	normalised	3.3235	3.2726	1.383	per 100 px
ring_attached_length_r30_px	network v2	317.565	262	197.219	px
ring_attached_length_r60_px	network v2	513.478	387	351.302	px
ring_length_w_r30_px	network v2	414.67	359.421	226.734	px
ring_length_w_r60_px	network v2	820.797	688.085	414.906	px
sholl_crossings_r10	network v2	9.4348	7	5.7432	count
sholl_crossings_r20	network v2	9	9	4.5227	count
sholl_crossings_r30	network v2	9	7	4.6221	count
loop_count	loops, report only	2.6087	1	3.7506	count
loop_edge_share	loops, report only	0.1232	0.0312	0.1463	share
ring_width_mean_r30_px	width	6.0954	6.0001	1.3892	px

Per field
----

measure	value	units
analysed_area_px2	715530	px^2
total_skeleton_length_px	31163	px
canalicular_length_density_per_px	0.0435523	px^-1
junction_count	865	count
junction_density_per_px2	0.00120889	px^-2
lacuna_count	31	count

lacunae_per_px2	4.332e-05	px^-2
skeleton_component_count	513	count
mean_component_length_px	60.7466	px
median_component_length_px	13	px
field_length_density_w_per_px	0.0489843	px^-1
field_density_without_flagged_per_px	0.0288425	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	394	px
unattached_component_count	21	count
canalicular_length_density_attached_per_px	0.0435523	px^-1
field_length_density_with_unattached_per_px	0.044103	px^-1
excluded_area_px2	232055	px^2
analysed_field_area_px2	816521	px^2
loop_count	116	count
loop_edge_share	0.1767	share
edge_count	2869	count

Canalicular width, the new measure

measure	value	units
width_mean_px	6.7523	px
width_median_px	6.3246	px
width_p10_px	2.8284	px
width_p90_px	10.198	px
width_pixels_used	22390	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	771	55.5328	20.5533	2.7019	0.929	0.9472	-1.5691	7.0195	631.551	True
2	7284	154.784	78.6772	1.9673	0.8612	0.6991	1.149	35.5404	964.419	True
3	6314	134.83	63.6497	2.1183	0.8816	0.9167	0.0507	178.336	997.655	True
4	5337	127.538	62.1114	2.0534	0.8734	0.801	0.3858	185.322	29.3642	True
5	4103	100.115	82.0021	1.2209	0.5737	0.6131	-0.8887	193.446	716.814	False
6	1981	84.786	33.0207	2.5677	0.921	0.8099	1.542	238.014	656.048	False
7	13774	208.681	129.902	1.6065	0.7826	0.5883	1.1124	248.226	868.396	False
8	653	48.9306	18.0087	2.7171	0.9298	0.8718	1.3396	244.608	191.13	False
9	439	24.8605	23.4655	1.0594	0.3303	0.8905	-1.2149	286.371	223.007	False
10	511	42.2275	16.9827	2.4865	0.9156	0.8705	0.9114	308.108	170.11	False
11	663	44.6399	20.5821	2.1689	0.8874	0.8511	1.5097	320.038	647.3	False
12	1055	56.5297	24.2615	2.33	0.9032	0.8843	1.0383	419.101	230.957	False
13	5004	114.836	64.0671	1.7924	0.8299	0.7737	1.4164	447.89	919.938	False
14	4433	89.0461	67.7113	1.3151	0.6494	0.9205	0.2586	458.37	993.331	True
15	10085	139.637	111.833	1.2486	0.5988	0.7317	0.8054	658.107	781.547	False
16	1893	74.136	39.3577	1.8836	0.8474	0.6949	-1.3345	688.738	268.372	False
17	532	36.1298	20.9704	1.7229	0.8143	0.8512	0.5415	713.359	881.316	False
18	1441	68.9456	33.0049	2.089	0.878	0.7238	-1.5587	747.817	126.852	False
19	3491	77.9644	65.3058	1.1938	0.5462	0.7932	-1.5218	773.022	646.082	False
20	988	66.6467	25.2664	2.6378	0.9254	0.6694	-1.3153	754.104	873.76	False
21	838	68.2276	23.1999	2.9409	0.9404	0.6315	0.0687	782.087	1002.4	False
22	8184	138.719	91.3085	1.5192	0.7528	0.6732	-1.4216	853.859	480.226	False

23	1147	76.2214	22.4254	3.3989	0.9557	0.7246	1.3411	869.654	382.98	False
24	1270	71.948	33.6183	2.1401	0.8841	0.6499	-1.4065	891.32	930.428	False
25	1577	97.9	24.8484	3.9399	0.9673	0.7152	-1.2598	906.939	130.226	False
26	957	43.0583	35.4718	1.2139	0.5669	0.6965	1.5597	917.457	535.305	False
27	1036	53.8662	28.9914	1.858	0.8428	0.7623	-1.4483	935.929	453.511	False
28	2879	109.14	44.0418	2.4781	0.915	0.6894	-1.4893	959.511	178.398	False
29	8395	147.679	83.9411	1.7593	0.8227	0.7501	1.451	988.219	760.234	True
30	600	40.3759	22.5899	1.7873	0.8288	0.8299	0.4308	962.445	1013.19	True
31	3679	107.549	49.0131	2.1943	0.8901	0.797	-1.5087	1004.05	371.105	True

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	2	154	466	720.044	50	14.4009	5.7841
2	8	260	521	522.73	32	16.3353	5.5936
3	5	243	319	280.35	16	17.5219	7.1131
4	9	365	807	416.534	28	14.8762	5.6611
5	9	385	614	933.637	67	13.9349	5.5198
6	3	132	303	65.2132	3	21.7377	7.0904
7	23	858	1290	1365.57	81	16.8589	7.3663
8	4	206	407	243.38	12	20.2816	3.2714
9	6	200	343	206.61	19	10.8742	5.9249
10	3	201	352	221.723	9	24.6359	5.9292
11	6	208	439	214.581	19	11.2937	6.6083
12	7	285	730	705.784	39	18.097	5.9149
13	7	327	825	553.002	30	18.4334	5.5882
14	4	100	229	50.9706	4	12.7426	5.8318
15	19	593	1162	1511.85	96	15.7485	5.7585
16	10	390	1015	1707.5	112	15.2455	6.3176
17	2	41	170	30.4853	3	10.1618	4.2847
18	8	403	1011	842.914	48	17.5607	6.6716
19	15	546	1252	1559.37	101	15.4393	7.4136
20	5	253	515	358.635	21	17.0778	5.8391
21	8	255	484	273.35	20	13.6675	6.1172
22	24	710	1277	1719.33	121	14.2093	8.515
23	7	266	453	758.943	58	13.0852	7.3521
24	12	413	904	1059.34	54	19.6173	6.0001
25	16	541	1106	1842.24	115	16.0194	7.8099
26	11	322	589	556.86	43	12.9502	2.8284
27	10	286	341	423.777	32	13.243	4.5701
28	17	658	1199	1314.83	91	14.4487	7.502
29	13	590	1213	1453.66	98	14.8332	6.7917
30	3	167	352	401.777	26	15.4529	4.4867
31	13	419	838	764.571	48	15.9286	7.0248

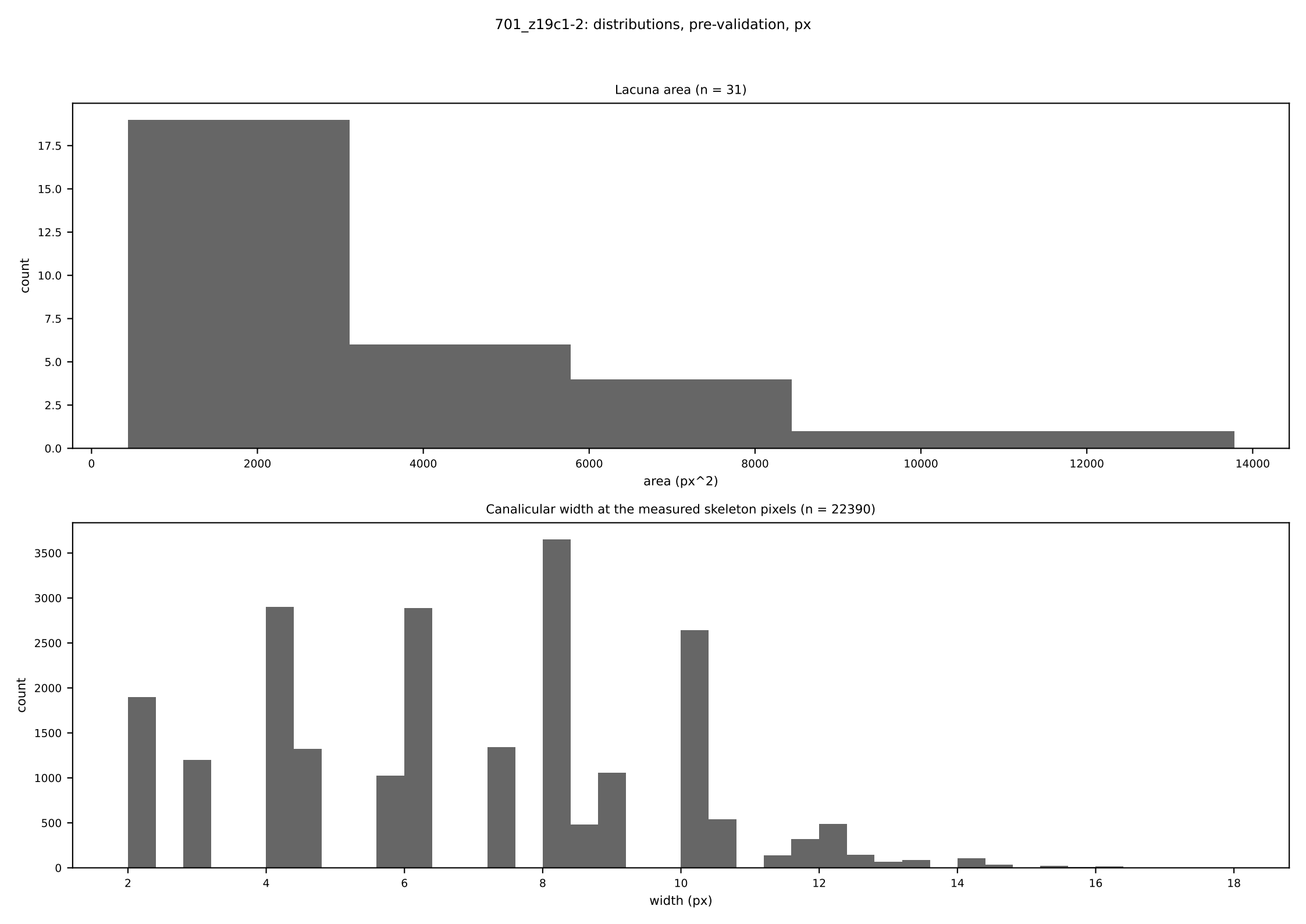
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	139.533	3684	10186	0.5367	0.5259	0.0418024	0.0457491	1.4334
2	723.844	8133	13577	0.5454	0.4636	0.0319685	0.0383737	1.1052
3	415.161	6608	8506	0.6083	0.5847	0.0367736	0.0375029	1.2044
4	459.973	9022	19412	0.687	0.6476	0.0404567	0.0415722	1.9566
5	687.3	9909	16328	1	1	0.0388536	0.0376041	1.3095
6	252.22	6062	10816	1	1	0.021775	0.028014	1.1894
7	1232.69	23194	37215	1	1	0.0369923	0.0346634	1.8658
8	122.225	4947	10475	1	1	0.0416414	0.0388544	3.2726
9	94.7107	3660	7794	1	1	0.0546448	0.0440082	6.3351

10	114.711	4934	9816	1	1	0.0407377	0.0358598	2.6153
11	134.61	6107	13723	1	1	0.0340593	0.0319901	4.4573
12	162.894	6964	18290	1	1	0.0409248	0.0399125	4.2973
13	403.931	9116	19283	1	0.9957	0.035871	0.0427838	1.733
14	311.338	4235	8482	0.6787	0.6401	0.0236127	0.0269984	1.2848
15	874.426	17209	32024	1	1	0.0344587	0.0362853	2.1729
16	260.777	9250	22508	1	1	0.0421622	0.0450951	3.8347
17	117.196	3262	6274	1	1	0.012569	0.027096	1.7065
18	308.818	8500	21039	1	1	0.0474118	0.0480536	2.5905
19	337.446	10732	22940	1	1	0.0508759	0.0545772	4.4452
20	200.25	5978	11822	1	1	0.0423219	0.0435629	2.4969
21	200.35	5923	11327	0.7826	0.6741	0.0430525	0.0427298	3.993
22	987.088	12715	21345	1	1	0.0558396	0.0598267	2.4314
23	217.421	5013	10115	1	1	0.053062	0.044785	3.2196
24	267.984	8463	17369	1	0.9865	0.0488007	0.0520468	4.4779
25	301.404	8091	17268	1	1	0.0668644	0.0640491	5.3085
26	255.806	4850	9732	1	1	0.0663917	0.060522	4.3001
27	199.622	4869	6483	1	1	0.058739	0.0525991	5.0095
28	503.269	10225	18917	1	0.9316	0.0643521	0.0633821	3.3779
29	834.04	12313	25028	0.7144	0.6685	0.0479168	0.0484657	1.5587
30	148.953	3707	6814	0.5954	0.519	0.0450499	0.0516583	2.0141
31	343.563	7295	12577	0.633	0.5959	0.0574366	0.0666296	3.7839

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	86	347	166.912	520.747	3	4	6
2	177	378	292.693	606.085	6	4	7
3	209	247	280.392	363.576	7	9	6
4	314	420	414.433	908.637	10	8	14
5	323	427	431.777	688.085	11	11	3
6	75	75	141.983	338.007	4	4	6
7	766	1071	984.548	1472.82	25	17	17
8	193	264	234.238	456.161	5	5	5
9	148	196	223.924	380.563	6	5	4
10	147	214	232.137	407.475	5	7	6
11	177	213	235.066	493.718	7	4	4
12	242	387	328.635	823.394	7	8	8
13	174	344	359.421	916.767	6	10	12
14	55	55	108.255	251.823	2	1	6
15	488	647	670.713	1314.9	17	16	11
16	256	583	434.777	1151.18	7	14	15
17	21	21	42.5563	194.995	1	0	6
18	333	747	448.363	1126.75	10	10	13
19	510	1004	606.647	1394.52	13	15	14
20	187	310	288.279	581.073	5	9	5
21	220	246	289.35	539.174	6	6	7
22	697	1190	797.399	1443.16	17	12	13
23	262	386	299.48	501.848	6	5	7
24	332	535	461.848	1020.85	9	10	15
25	537	1066	602.576	1241.02	19	13	10
26	315	484	363.907	663.286	7	8	6
27	276	317	322.421	385.291	9	4	3
28	625	1083	737.357	1343.29	15	14	17
29	473	848	657.801	1358.64	11	11	15
30	121	241	190.752	395.948	3	6	9
31	340	554	467.404	930.997	8	10	14


701_z19c1-2: distributions, pre-validation, px
Lacuna area (n = 31)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 22390)
count
width (px)